

Lyapunov Functions (Meiss 4.6).

Recall: Suppose x^* is an equilibrium of φ_t in $\mathbb{R}^n$. Then

x^* is Lyapunov stable $\Leftrightarrow$ for every neighbourhood N of x^*
 $\exists M \subseteq N$ s.t. $\varphi_t(x) \in N$ for all $t \geq 0, x \in M$.

x^* is asymptotically stable $\Leftrightarrow$ it is Lyapunov stable and
 $\varphi_t(x) \rightarrow x^*$ as $t \rightarrow \infty$ for all
 x close to x^* .

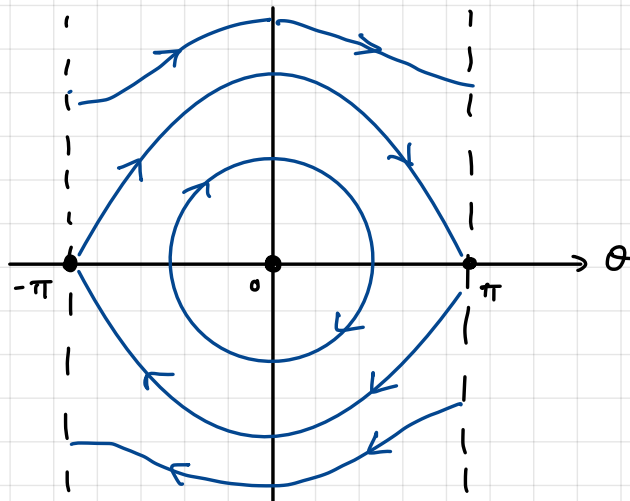
Example: We can consider two model equations for a pendulum.

(1) Undamped Pendulum

$$\ddot{\theta} = -\sin \theta$$

$$\Rightarrow \begin{cases} \dot{\theta} = y \\ \dot{y} = -\sin \theta \end{cases}$$

The origin is Lyapunov stable but not asymptotically stable.

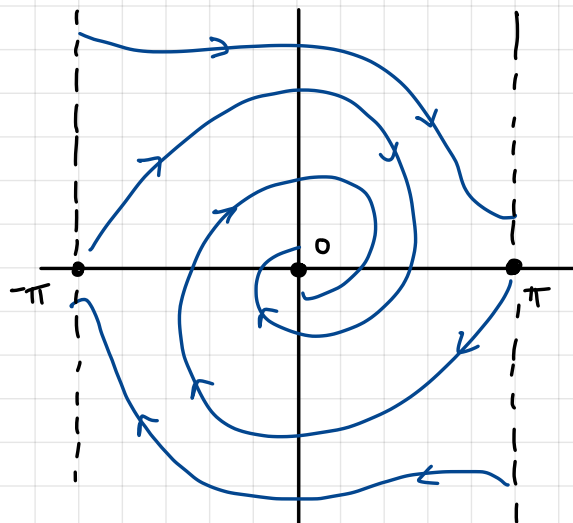


(2) Damped Pendulum.

$$\ddot{\theta} = -\sin \theta - \beta \dot{\theta}, \quad \beta > 0$$

$$\Rightarrow \begin{cases} \dot{\theta} = y \\ \dot{y} = -\sin \theta - \beta y \end{cases}$$

The origin becomes asymptotically stable.



But how do you show a fixed point is Lyapunov stable?

Nonlinear ODEs are not explicitly solvable, and Lyapunov stability is usually used when the linearization is inconclusive, as is the case for the center fixed point in (1) above.

The examples above suggest a possible method.

The undamped pendulum is conservative ie it has an energy which remains constant on orbits. Define:

$$E(x,y) = \frac{1}{2}y^2 - \alpha \cos x$$

$$\Rightarrow \frac{dE}{dt} = y\dot{y} + \alpha \sin x \dot{x} = -\alpha y \sin x + \alpha y \sin x = 0.$$

As such, the orbits of the pendulum are contained in the level sets of E . Hence, given a neighbourhood N of x^* , we can choose M by making sure all the level sets intersecting M stay inside N . This is possible for $(x,y)=(0,0)$ as this is a local minimum of E .

This also works for the damped pendulum. Using the same function E :

$$\begin{aligned} \frac{dE}{dt} &= y\dot{y} + \alpha \sin x \dot{x} \\ &= y(-\alpha \sin x - \beta y) + \alpha y \sin x \\ &= -\beta y^2 \leq 0 \end{aligned}$$

In this case the energy is not constant but always decreases. So once inside a level set near zero, orbits must remain there. Moreover, the decreasing property together with the fact $x^*=0$ is the only local minimum of E gives some hope we can also get asymptotic stability from this method.

This is the idea of Lyapunov functions: constrain the dynamics by the level sets of a function.

Def: A continuous function $L: \mathbb{R}^n \rightarrow \mathbb{R}$ is a (strong) Lyapunov function for an equilibrium x^* of φ_t on $\mathbb{R}^n$ if $\exists$ open set U of x^* s.t.

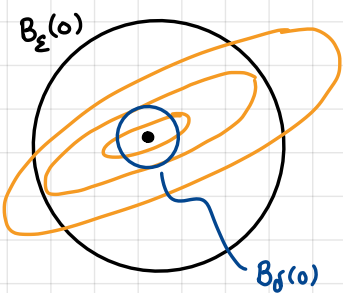
- (i) $L(x^*) = 0$
- (ii) $L(x) > 0$ for all $x \neq x^*$
- (iii) $L(\varphi_t(x)) < L(x) \quad \forall x \in U \setminus \{x^*\}$ and $t > 0$.

L is instead a weak Lyapunov function if (iii) is replaced by
(iv) $L(\varphi_t(x)) \leq L(x) \quad \forall x \in U$ and $t > 0$.

Thm: (Lyapunov Function)

Let x^* be an equilibrium of φ_t . If L is a weak Lyapunov function for x^* , x^* is Lyapunov stable. If L is a strong Lyapunov function for x^* , x^* is asymptotically stable.

Proof: Without loss choose $x^* = 0$, and let L be a weak Lyapunov function on a neighbourhood U . Then, for any $\varepsilon > 0$, $B_\varepsilon(0) \subset U$ and $m = \min\{L(x) : |x| = \varepsilon\}$ exists and is positive.



By continuity, $L(x) \rightarrow 0$ as $x \rightarrow 0$, so we can find a $\delta > 0$ s.t. $L(x) < m$ for all $x \in B_\delta(0)$.

Moreover, L has the property that
 $L(\varphi_t(x)) < L(x) < m$ for all $x \in B_\delta(0)$.

ie. $\varphi_t(x)$ cannot cross the boundary of $B_\varepsilon(0)$.

We conclude x^* is Lyapunov stable.

Suppose in addition L is a strong Lyapunov function.
We want to show that for any $x \in B_\delta(0)$, $\varphi_t(x) \rightarrow 0$.

Since we have already shown $\varphi_t(x) \in B_\varepsilon(0)$, it is bounded
and so $\exists \{t_n\}$ s.t. $t_n \rightarrow \infty$ and $\varphi_{t_n}(x)$ converges as $i \rightarrow \infty$.

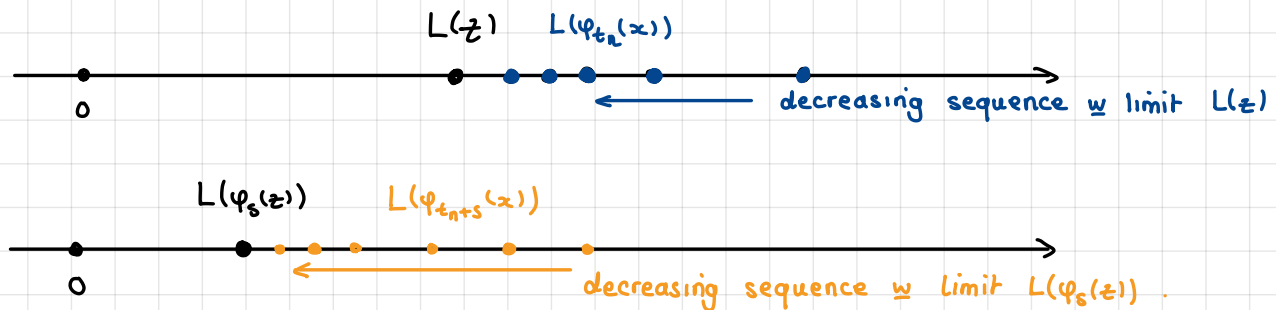
(by Bolzano-Weierstrass)

Suppose by contradiction $\varphi_{t_n}(x) \rightarrow z \neq 0$. Then we can construct two monotone decreasing sequences:

$$L(\varphi_{t_1}(x)) > L(\varphi_{t_2}(x)) > \dots > L(\varphi_{t_n}(x)) > \dots \quad \text{and} \quad \lim_{n \rightarrow \infty} L(\varphi_{t_n}(x)) = L(z)$$

$$L(\varphi_{t_1+s}(x)) > L(\varphi_{t_2+s}(x)) > \dots > L(\varphi_{t_n+s}(x)) > \dots \quad \text{and} \quad \lim_{n \rightarrow \infty} L(\varphi_{t_n+s}(x)) = L(\varphi_s(z))$$

for $s > 0$ and the limits come from the continuity of L and φ .



But this is a contradiction as we can find $t_m > t_n + s$ for which

$$L(z) < L(\varphi_{t_m}) < L(\varphi_{t_n+s}) < L(z) \quad \text{⚡}$$

□

Remark:

- Lyapunov functions are a powerful tool, but can be very hard to find. In the pendulum example you can find the energy using physical intuition.

Beyond some very simple situations, the only time there is a good way to find a Lyapunov function is when your system has additional structure.

Eg. conservative, Hamiltonian or gradient systems.

Example:

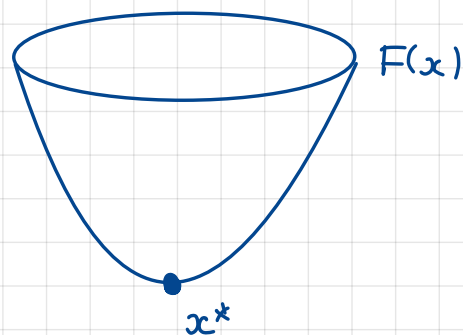
$$(1) \quad \dot{x} = -\nabla F(x)$$

If x^* is a local minimum of $F(x)$,
 $-\nabla F(x)$ points downhill.

We can take $L(x) = F(x)$. Then

$$\begin{aligned} \frac{d}{dt} (L(x(t))) &= \nabla F(x(t)) \cdot \dot{x}(t) \\ &= -|\nabla F(x(t))|^2 \leq 0 \end{aligned}$$

with equality only if $x = x^*$. So L is a strong Lyapunov-function on a neighbourhood of x^* .



$$(2) \quad \ddot{x} = -\nabla F \Rightarrow \begin{cases} \dot{x} = y \\ \dot{y} = -\nabla F \end{cases}$$

These are mechanical equations and have conserved energy:

$$E(x, y) = \frac{1}{2} |y|^2 + F(x) .$$

which makes a (weak) Lyapunov function.

$$(3) \text{ Hamiltonian System. } \begin{cases} \dot{x} = \frac{\partial H}{\partial y} \\ \dot{y} = -\frac{\partial H}{\partial x} \end{cases}$$

Again there exists a conserved energy: H .

$$\frac{dH}{dt} = \frac{\partial H}{\partial x} \cdot \dot{x} + \frac{\partial H}{\partial y} \cdot \dot{y} = \frac{\partial H}{\partial x} \frac{\partial H}{\partial y} - \frac{\partial H}{\partial y} \frac{\partial H}{\partial x} = 0 .$$

So if (x^*, y^*) is a local minimum of H ,
 $L(x, y) = H(x, y) - H(x^*, y^*)$ is a (weak) Lyapunov function.

Remark:

Also in the damped pendulum case we cannot conclude asymptotic stability as E does not decrease when $y=0$. However, by the phase portrait, we expect 0 is asymptotically stable.

We can sometimes strengthen the result for a weak Lyapunov function, assuming we can understand the set $\{dL/dx = 0\}$.

Thm: (La Salle's Invariance Principle)

Suppose x^* is an equilibrium for $\dot{x} = f(x)$ and that L is a weak Lyapunov function for x^* on a compact, forward-invariant neighbourhood U of x^* .

Let $Z = \{x \in U : \frac{dL}{dt} = 0\}$ be the set where L is not decreasing.

Then if x^* is the largest forward invariant subset of Z , it is asymptotically stable and attracts every point in U .

Question: How does this work in the case of the pendulum equations above?

Lyapunov functions can also be used to prove asymptotic stability of fixed points with stable spectrum.

Lemma: An asymptotically stable linear system always has a Lyapunov function of the form $L = x^T S x$, S symmetric.

Proof: Let $\dot{x} = Ax$ where all eigenvalues of A have negative real parts.

Then suppose L is a function $L = x^T S x$ for some matrix to be determined. Then assuming S is symmetric

$$\begin{aligned}\frac{dL(x(t))}{dt} &= \frac{d}{dt} (x^T S x) = \dot{x}^T S x + x^T S \dot{x} \\ &= x^T A^T S x + x^T S A x \\ &= x^T (A^T S + S A) x\end{aligned}$$

If we could find S s.t. $AS + SA = -I$ then we would have
 $\frac{dL}{dt} = -|x|^2 \leq 0$ with equality only at $x=0$.

If such an S is also positive definite then $L(x) \geq 0$ and L is a strong Lyapunov function.

Now: $A^T S + SA = -I$

$$\Leftrightarrow e^{A^T t} (A^T S + SA) e^{At} = -e^{A^T t} e^{At}$$

$$A^T e^{A^T t} S e^{At} + e^{A^T t} S A e^{At} =$$

$$\frac{d}{dt} (e^{A^T t} S e^{At}) =$$

$$\Rightarrow e^{A^T t} S e^{At} = C + \int_0^t -e^{A^T \tau} e^{A \tau} d\tau$$
$$= \int_t^\infty e^{A^T \tau} e^{A \tau} d\tau \quad (\text{Choose } C = \int_0^\infty e^{A^T \tau} e^{A \tau} d\tau)$$

Taking $t=0 \Rightarrow S = \int_0^\infty e^{A^T \tau} e^{A \tau} d\tau.$

Note the matrix S is only defined if A has eigenvalues with negative real part since then

$$\|e^{At}\| \leq e^{\alpha t} \quad \text{where } \alpha = \max_n \operatorname{Re} \lambda_n < 0$$

Hence, $\|e^{A^T \tau} e^{A \tau}\| \leq e^{2\alpha \tau}$ and the improper integral is well-defined.

Claim: S is symmetric positive definite.

$$\cdot S^T = \left(\int_0^\infty e^{A^T \tau} e^{A \tau} d\tau \right)^T = \int_0^\infty (e^{A \tau})^T (e^{A^T \tau})^T d\tau = S.$$

$$\cdot \text{For any } x \in \mathbb{R}^n \setminus \{0\} \quad x^T S x = \int_0^\infty \underbrace{x^T e^{-A^T \tau} e^{A \tau} x}_{> 0} d\tau > 0.$$

We conclude that L is a strong Lyapunov function for the origin on all of $\mathbb{R}^n$. $\square$

This result can then be extended to nonlinear systems with spectrally stable fixed points as well.

Thm: Consider $\dot{x} = Ax + g(x)$ with an equilibrium at 0 and $g(x) = O(|x|^2)$.

If A has eigenvalues only in the left half plane, there is an open set U of zero where $L = x^T S x$ is a strong Lyapunov function.

Proof: We construct L as in the previous lemma.

$$\begin{aligned}\text{Then } \frac{dL(x(t))}{dt} &= \dot{x}^T S x + x^T S \dot{x} \\ &= (Ax + g(x))^T S x + x^T S (Ax + g(x)) \\ &= x^T A^T S x + x^T S A x + g(x)^T S x + x^T S g(x) \\ &= -x^T x + 2x^T S g(x)\end{aligned}$$

$$\text{But } |x^T S g(x)| \leq |x| |g(x)| \leq K |x|^3.$$

Hence, for small enough x we can guarantee $2x^T S g(x)$ is smaller than $x^T x = |x|^2$.

ie. L is again a strong Lyapunov function. $\square$